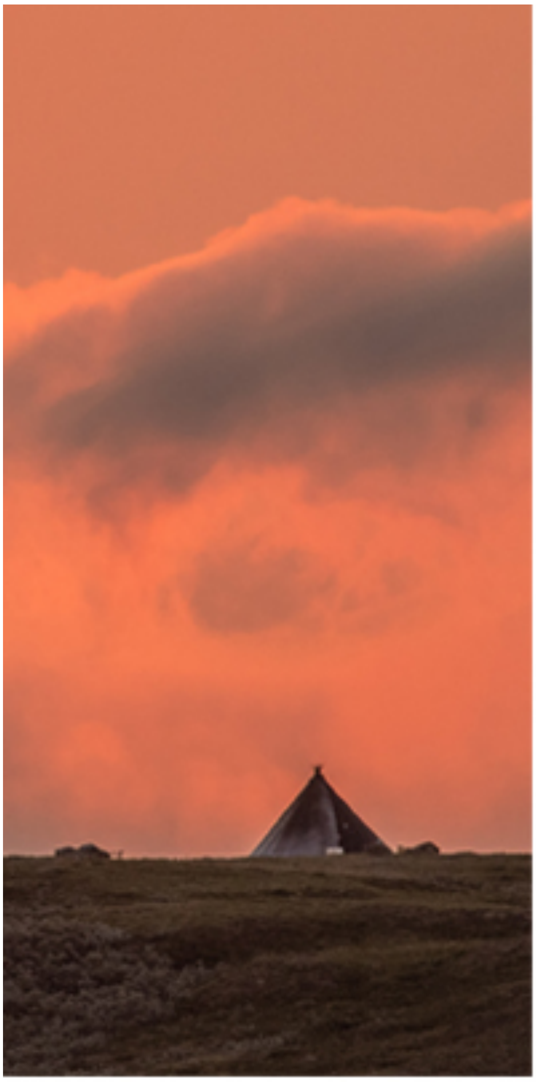
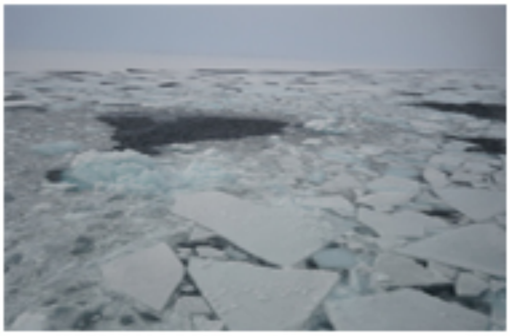


CHARTER Working Paper 3

The Future of Reindeer Husbandry: Surprises & Dreams

CHARTER

Sirpa Rasmus, Simo Sarkki, Antti-Juhani Pekkarinen, Mikko Jokinen, Ilona Mettiäinen, Laura Post, Taru Rikkonen, Jaana Sorvali, Tuulia Väärälä.



In this report, we present the outcomes of the ”Future of Reindeer Husbandry: Surprises and Dreams” workshop. The workshop was organized in Inari in connection with the Reindeer and Fish Science Days (*Poro- ja kalapäivät*) on 25 August 2022. The objective of the interactive workshop was to examine and discuss the current state and future developments of reindeer husbandry and how reindeer herders can prepare for surprising future developments.

The workshop was organized by the projects POVAUS and CHARTER of the University of Lapland’s Arctic Centre and the Natural Resources Centre (LUKE) in collaboration with the CLIMINI and POMURI projects. Some organizers were from the Lapland University of Applied Sciences, as well. The themes addressed support the ongoing work of the Future of Reindeer Husbandry *(Porotalouden tulevaisuus)* working group set by the Ministry of Agriculture and Forestry of Finland.

A longer version of this report in Finnish is available on the CHARTER project website: tinyurl.com/uvua2see

The Future of Reindeer Husbandry: Surprises and Dreams
A Workshop Summary Report

Reindeer husbandry is an important northern livelihood and a culturally significant way of life for both Sámi and non-Sámi Finns practicing reindeer herding. Possibilities of ensuring continuity of the livelihood are currently being affected by global developments such as climate change and its mitigation, the green transformation, the war in Ukraine, and the global pandemic. In addition to assessing and preparing for the impacts of these, it is important to reflect on the developments that are taking place within the livelihood itself as well as the ways in which the existing legislation and administrative practices enable or slow down achievement of the goals of reindeer herding and reindeer husbandry.

Figure 1 Work at various stages of the workshop.

In this report, we present the outcomes of the ”Future of reindeer husbandry: surprises and dreams” workshop. The workshop was organized in Inari in connection with the Reindeer and Fish Science Days (*Poro- ja kalapäivät*) on 25 August 2022 (Figure 1). The objective of the interactive workshop was to examine and discuss the current state and future developments of reindeer husbandry and how reindeer herders can prepare for surprising future developments.

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Participants and method

A total of 36 people, of which 9 were organizers, participated in the workshop. The proper participants represented seven reindeer herding cooperatives and the Municipality of Inari, the Regional State Administrative Agency for Lapland, the Lapland University of Applied Sciences, the Natural Resources Centre (LUKE), the Central Union of Agricultural Producers and Forest Owners (MTK), Reindeer Herders' Association, the Sámi Parliament (Sámediggi), the Sámi Reindeer Herding Cooperatives (Sámi bálgosat rs) and Suoma Boazosámit rs. The participants were divided into groups and seated at four tables.

Figure 2. An example of an operational environment of reindeer herding built during the workshop.

The method used was based on an interactive futures tool developed by the organizers, as well as discussion informed by it. The participants were asked to:

1) Build a conceptual map of the current operational environment of reindeer husbandry. For this purpose, the participants received a deck of cards from which they were asked to select cards that were representative of the elements of the operational environment of reindeer husbandry. These had been divided into four general themes: 1) "Pastures and land use", 2) "Reindeer herding and reindeer herders", 3) "Climate, the environment, reindeer" and 4) "Economy, market, society and governance". The conceptual map consisted of the elements represented by the cards and the interactive relations between them.

2) Reflect on how global developments and surprises affect the operational environment of reindeer herding and how it is possible to prepare for change. The developments selected for the workshop were "accelerating climate change", "strong climate change mitigation and green transformation", "continuing state of war and soaring expenses" and "time of pandemics".

3) Present dreams and goals related to reindeer herding and to identify possible critical factors affecting their achievement in the previously built conceptual map representing the operational environment of reindeer herding (Figure 2).

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The Current state of reindeer herding: key factors of the operational environment

Participants at each table were given 48 cards representing the elements of the operational environment of reindeer herding. Each table chose about 20 of these. This was also an act of prioritizing that made visible the things that the participants found the most important in describing the current state of the operational environment of reindeer herding. The participants also had blank cards at their disposal and they used them to add the following elements to the conceptual maps: “forest nature”, “grazing peace”, “the many impacts of tourism”, “reindeer herding workforce retention”, “legal practices”, “the compromised position of reindeer herding in legislation”. Two tables introduced “climate change” as part of the operational environment already at this stage, although more in-depth discussion on the subject was planned for and took place at later stages of the workshop. Table 1 presents only the cards that every table chose as the most important elements of the operational environment of reindeer herding. A more detailed table showing all the elements examined is presented in the full report (in Finnish).

The round in which the card was picked	1.	2.	3.
Land-use planning	3	1	
Pasture rotation	3	1	
Traditional knowledge, competence, education	2	1	1
Cooperation, communality, interaction	2	1	1
Profitability and expenses	2	1	1
Winter pastures (quantity, quality, vegetation, usability)	2	1	1
National legislation	1	2	1
Reindeer population and reindeer well-being	1	3	
Winter weather and snow conditions	1	3	
Reindeer herder well-being and coping		3	1

Table 1. The titles of the cards that were chosen at every table.

Surprises and global drivers

To prepare for the future, it is necessary to examine how global forces of change (referred to as *global drivers*) may impact reindeer herding. In this workshop, the participants started by choosing cards and building a map of the operational environment of reindeer herding, also showing the relations between the cards chosen (Figure 2). During the second stage of the workshop, the participants were asked to discuss how the four global drivers impact the operational environment of reindeer herding, what can be done to prepare for the anticipated change and by whom. The developments, of which each table addressed one, had been chosen by the workshop organizers and they were: "accelerating climate change", "strong climate change mitigation and green transformation", "continuing state of war and soaring expenses" and "time of pandemics". Table 2 lists some of the impacts identified and ways of preparing for them that emerged in the discussions.

The developments listed in Table 2 can no longer be referred to as surprises. According to the experiences of reindeer herders, visible impacts of climate change are not a matter of the distant future, but part of the reality of reindeer herding today. Two recent winters (2019–2020 and 2021–2022), which were particularly challenging from the perspective of reindeer herding, can be mentioned as an example.

Global forces of change can also lead to a surprising combination of effects. At worst, global attempts to mitigate climate change will fail despite increased adoption of land use solutions that meet the requirements of the green transformation in the reindeer management area. Geopolitical tensions are ongoing, state of war in the neighbouring regions continues, and pandemics are here to stay. The number of international tourists is declining, and making a living as a multi-industry entrepreneur becomes increasingly challenging. Increased frequency of difficult winter conditions leads to increased demand for supplementary winter feeding for reindeer, and simultaneously, prices for forage and fuel are rising. Taken together, the above may send reindeer herding on a downward spiral that puts the sustainability and continuity of the livelihood at risk. In addition, smaller-scale surprises may be experienced at the local level.

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Future challenges	Ways of preparing
Accelerating climate change: - Increased frequency of extreme weather phenomena such as heat periods and storms - Unprecedented snow conditions such as very deep snow or a very short snow period - Increased frequency of freezing of pastures and increased mould growth on them, icy snow layers - Dry and hot summers - Reindeer diseases, parasites and insect harassment - Changes in the growth of lichen, arboreal lichen and mushrooms	- A rapid response catastrophe fund for adapting to climate change, financed by e.g. municipalities and the state; - A compensation system for adapting to climate change; - A monitoring system for predicting difficult conditions and preparing for them; - Emergency feeding of reindeer and external help in times of crisis (e.g. the military as in Sweden and Norway); - Resources for reindeer herders to adapt and cope; - Forestry solutions that are beneficial from the perspective of reindeer herding.
Climate change mitigation and green transformation: - Increase of renewable energy production forms in the reindeer management area (e.g. wind energy) - A growing number of mines, as demand for minerals required for battery technology increases? - Support by conservation areas to reindeer herding (if reindeer are not excluded from these areas); - Pressure to reduce using petrol-driven motor vehicles; - Decrease in international air travel; - Possible decrease in the number of reindeer herders; - Decrease in profitability of reindeer herding as a result of other land use activities?	- Combining technological solutions (e.g. drones) and traditional means (e.g. reindeer herding dogs); - Lawsuits to secure recognition of reindeer herders' rights; - Subsidies, compensation and compensation for damage to ensure justness; - A cultural / green certificate for reindeer herding; - Forestry measures that treat forests as carbon sinks and reindeer pastures simultaneously; - Reducing follis fuel consumption, which cleans the air and may thus provide better conditions for arboreal lichen growth?
Continuing state of war and soaring expenses: - Increase in forage prices; - Increase in fuel prices; - ”A New Chernobyl” in Ukraine? - Demands for increasing self-sufficiency both in food and energy production.	- Focus on the significance of primary production is beneficial for reindeer herding (e.g. demand and price for meat)

Pandemics - Seeking new sales channels for meat; - Large buyers buy at lower prices; - A very low number of tourists; - Difficulties in organizing work within the family if a family member's ability to work is compromised; - Changes to work tasks performed together and organizing meetings - Future risks of diseases in animals that can be transmitted to humans (zoonoses)?	- Limiting entry of outsiders to reindeer round-ups - Keeping one's vaccination status up to date; - Observing good hygiene practice; - Acquiring the skills required to take the digital leap - Improving occupational health care of reindeer herders - Self-sufficiency in food for reindeer herders - Living in remote areas and the contentment it brings - Improving the ability to organize and distribution of tasks in times of crisis

Dreams: "Reindeer herding without crises" or "Maintaining reindeer herding as a sound livelihood despite crises?"

During the final stage of the workshop, the participants wrote down their dreams and goals related to reindeer herding and linked these with the most critical related points of the operational environment of reindeer herding as represented in the conceptual maps built at the beginning of the workshop. In addition, the participants reflected on what would have to change in order for their dreams to become reality. These discussions are summarized in Figure 3 using the "Three Horizons" visualization method. Horizon 1, marked in red, focuses on the current state of issues that should be changed. Horizon 2, marked in green, represents things that enable change for the better and progress towards the achievement of the goals. Horizon 3, marked in blue, represents long-term dreams related to reindeer herding.

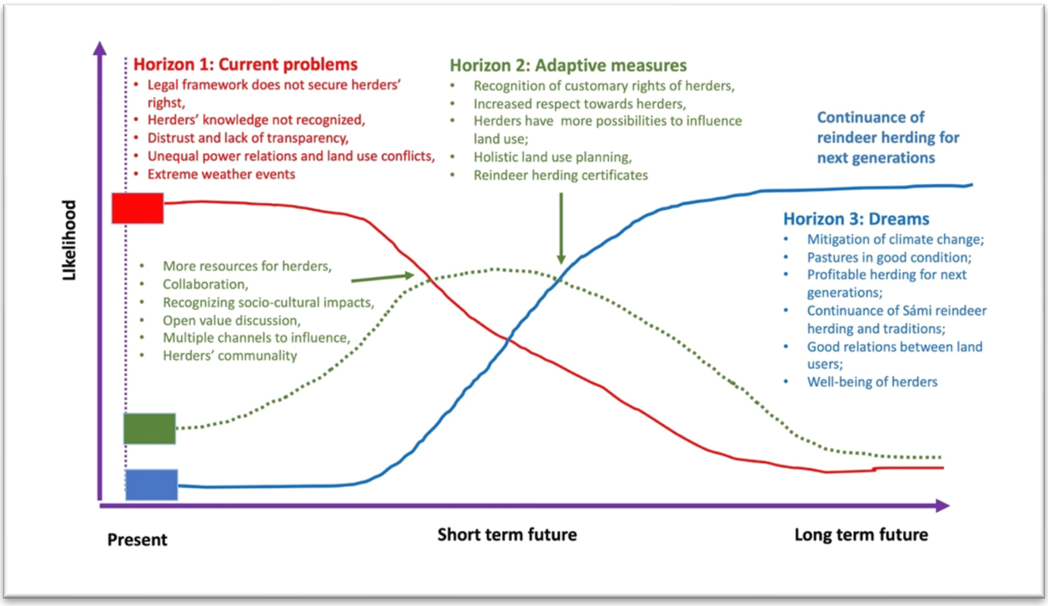


Figure 3: How to enable progress towards realization of the dreams related to reindeer herding? Here the "Three Horizons" approach is used, where the colour red represents the problematic issues experienced at present, the colour green stands for positive transformative factors that enable change, and the colour blue indicates desirable futures.

Recommendations based on the workshop for work on futures of reindeer husbandry

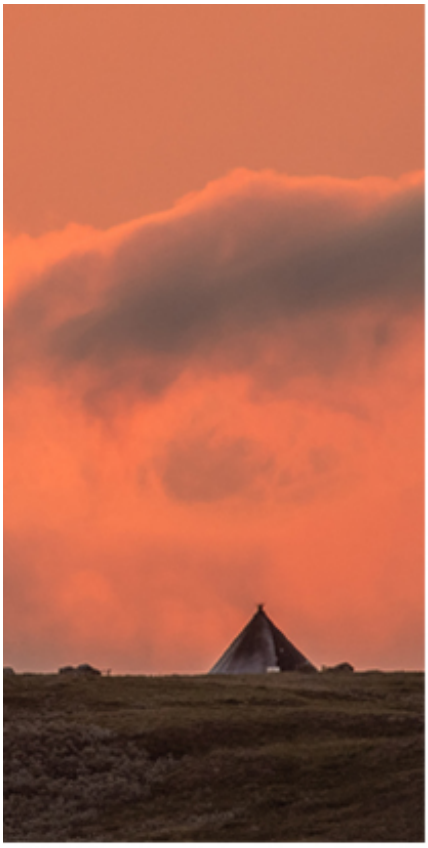
1) Climate change is not the only global force of change shaping the current state and future of reindeer herding. In addition to changes occurring in the natural environment, strong climate change mitigation (referred to as the green transformation or the green transition), ongoing geopolitical tensions and the continuing state of war in the neighbouring regions, as well as the many long-term impacts of the COVID-19 pandemic create pressures for change for reindeer herding. In addition to considering factors related to climate change, work on futures of reindeer husbandry should also consider combined effects of the global drivers and prepare for the anticipated change (Table 2).

2) Then again, climate change is already happening at present and it is affecting the reindeer management area, but what reindeer husbandry still lacks is 'a climate roadmap' or an adaptation plan that would guide the livelihood in its efforts towards adapting to climate change. Acts of coping and adaptation take place in the everyday life and work of reindeer herders. Sustainable adaptation requires administrative support, a long-term plan, and ongoing monitoring of the changing conditions.

3) When discussing realization of dreams related to reindeer herding, ”land use planning” was the most frequently mentioned as the most critical element of the operational environment. Land use planning is also integrally linked with the impacts involving several of the global drivers (Table 2). Essentially, land use planning also has an impact on grazing peace, and through that, on the well-being of reindeer and reindeer herders. These are inextricably connected with each other. The status of reindeer herding in land use planning could be enhanced for example by 1) appointing for each reindeer herding cooperative a person responsible for maintaining the relationship between reindeer herding and other land users; 2) engaging in overall planning in the area of the reindeer herding cooperative that enables considering the combined effects of the diverse land uses in the area on reindeer herding; and 3) paying greater attention to the socio-cultural impacts of land use.

4) Transformative factors enabling progress towards realization of dreams and achievement of goals related to reindeer herding should be supported through administration. Shaping change for desired futures does not only mean adding positive things but also mitigating – or changing – negative ones. The current state of the operational environment of reindeer herding includes elements that require change if reindeer husbandry is to continue to thrive as a livelihood (Figure 3).

Through supporting positive transformative factors reindeer herding could have a future where it can “exist without crises”, or, at least continue to thrive despite crises. Global forces of change have an impact on reindeer herding, but they do not determine its future. By preparing for change and by supporting positive and desired developments reindeer herding can achieve sustainability, continuity and well-being in a world in turmoil.

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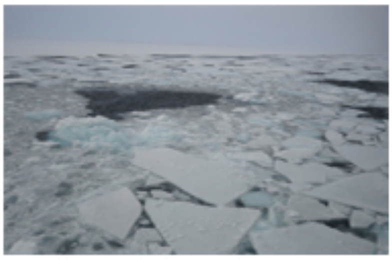


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The CHARTER project name is derived from the research project title: Drivers and Feedbacks of Changes in Arctic Terrestrial Biodiversity. CHARTER is an ambitious effort to advance the adaptive capacity of Arctic communities to climatic and biodiversity changes through state-of-the-art synthesis based on thorough data collection, analysis and modelling of Arctic change with major socio-economic implications and feedbacks.

CHARTER involves 21 research institutions across 9 countries. The 4 year, 5.9M Euro project is managed by the Arctic Centre at the University of Lapland, Finland and is funded by the EU Horizon 2020 Research and Innovations Programme (Grant #869471).

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